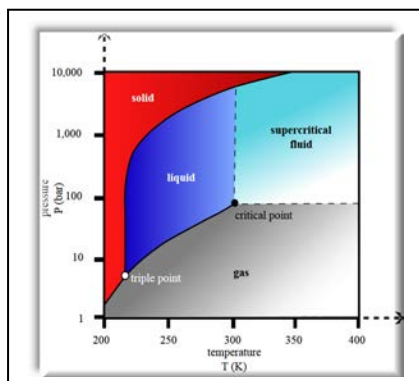


UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering



College: UIC COLLEGE OF ENGINEERING
Department: UIC Department of Chemical Engineering
Course: CHE 301: Chemical Engineering Thermodynamics
Time and Place: SES 138 MW: 03:00 P.M. – 04:15 P.M. F: 03:00 P.M. – 03:50 P.M.
Instructor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu
Cell Phone: (630) 696-0934

COURSE OBJECTIVES: This course is designed to extend the student's acquired knowledge and understanding of the first and second law of thermodynamics to the most critical areas of Chemical Engineering Thermodynamics and Chemical Reaction Equilibrium. These areas

encompass properties of pure fluids and fluids in homogeneous solutions, single and multicomponent phase equilibrium, and chemical equilibrium. The knowledge of this material will provide a strong foundation for all subsequent courses in chemical engineering and contribute towards understanding of chemical processes. By the end of the course the students should be able to:

- ✿ Understand the basic concepts of Internal Energy, Gibbs Energy, Entropy and Enthalpy.
- ✿ Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.
- ✿ Understand the theory of thermodynamics of ideal and non-ideal solutions
- ✿ Understand the applicability and use of various equations of state in describing the thermodynamic state of pure fluids and mixtures thereof.
- ✿ Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.
- ✿ Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

COURSE PREREQUISITES: The following are key requirements for a successful completion of this course.

- **Integral and Differential Calculus.**
- **Multivariable Differential Calculus.**
- **Familiarity with a Programming Language.**
- **CHE Department Prerequisites: CHE 201, CHE 205 and CHE 210**

SOFTWARE: Students must download [FREE of charge] from the UIC Blackboard the software programming package [Athena Visual Studio](#) and the EXCEL based Thermodynamics Calculator [CHE301 Calculator.xlsm](#)

RECOMMENDED TEXTBOOK AND NOTES:

1. Michael Caracotsios, *Chemical Engineering Thermodynamics Lecture Notes*, will be distributed to all students free of charge via the UIC Blackboard, 2020
2. Themis Matsoukas, *Fundamentals of Chemical Engineering Thermodynamics*, Prentice Hall, 2013.

ADDITIONAL REFERENCES:

1. J. M. Smith, H. C. Van Ness and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th Edition, McGraw-Hill, 2005.
2. S. I. Sandler, *Chemical and Engineering Thermodynamics*, 3rd Edition, John Wiley & Sons, Inc., 1999.

CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS COURSE OUTLINE

1. REVIEW AND BACKGROUND

[CHAPTER 1 IN TEXTBOOK]

- Introduction to Thermodynamics in Chemical Engineering
- Elements of Differential and Integral Calculus
- Numerical Methods for Algebraic and Differential Equations
- Mass and Energy Balances in the Context of Thermodynamics
- Tutorials on Programming with VBA and Athena Visual Studio

2. THERMODYNAMIC PROPERTIES OF SINGLE-COMPONENT FLUIDS

[CHAPTER 7 IN TEXTBOOK]

- The PVT Behavior of Single-component Fluids
- The Gibbs Phase Rule for Single-component Fluids
- The Ideal Gas Behavior
- Cubic and Virial Equations of State
- Fugacity and Fugacity Coefficients from Equations of State
- Thermodynamic Properties of Single-Phase and Two-Phase Systems

3. THERMODYNAMIC PROPERTIES OF MULTI-COMPONENT MIXTURES

[CHAPTERS 9, 10, 11, 12 IN TEXTBOOK]

- Partial Molar Properties
- Excess and Residual Thermal Properties
- The Gibbs Phase Rule for Multi-component Mixtures
- The PVT Behavior of Multi-component Mixtures
- Cubic Equations of State and Mixing Rules
- Fugacity and Fugacity Coefficients from Equations of State
- Liquid Activity Coefficients for Non-ideal Solutions
- Thermodynamic Properties of Single-Phase and Two-Phase Systems

4. MULTICOMPONENT PHASE EQUILIBRIUM

[CHAPTER 12 IN TEXTBOOK]

- Criteria for Phase Equilibrium
- Vapor-Liquid Equilibrium: The Fundamental Problem
- Flash Calculations: PT, PH, PV, TH, TV and HV Calculations
- VLE Data Analysis: Activity Coefficients from Experimental Data
- Liquid-Liquid Equilibrium: The Fundamental Problem ^{NEW}

5. CHEMICAL REACTION EQUILIBRIUM

[CHAPTER 14 IN TEXTBOOK]

- Criteria for Equilibrium in Chemical Reactions
- Gibbs Free Energy and Equilibrium Constants
- Evaluation of Equilibrium Constants from Thermodynamic Data
- Calculation of Equilibrium Conversion for Single and Multi-reaction Systems

ACADEMIC INTEGRITY:

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy:

<https://dos.uic.edu/conductforstudents.shtml>

RELIGIOUS HOLIDAYS:

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

The University Holidays and the Religious Observances calendar can be found online at the following website: <http://oe.uic.edu/religious-calendar/>. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY:

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Partial credit can also be requested in cases where there is an obvious (indisputable) error made by the Professor or the Teaching Assistant.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class:

Not allowed unless explicitly stated

Repeat Offenders will be asked to leave the classroom

Cheating/Copying: The entire class gets penalized after the first warning

Homework contribution to the final grade is substantially reduced

Perpetrators will receive zero points in their homework

Late Homework and Disputes:

Late Homework will not be accepted without prior arrangements with your Professor in person

If there are Homework and Exam score disputes they will be resolved only with the Professor.